

Portfolio

AI & Deep Learning Projects

No.	Project	Tech Stack	Description	 Hardness (1–10)
1	Object Tracker with RNN	YOLOv8, Kalman Filter, RNN	Combines spatial detection and temporal modeling for smooth tracking.	9
2	DeepLabV3 Image Segmentation Tool	PyTorch, DeepLabV3	Foreground-background segmentation with overlay support.	8
3	Anomalib PatchCore Classifier	PyTorch, Anomalib	Unsupervised anomaly detection using PatchCore model.	9
4	Facial Expression Classifier	MediaPipe, OpenCV	Real-time emotion detection from live camera feed.	6
5	Driver Drowsiness Detection	Dlib, OpenCV, EAR	Fatigue detection via facial landmarks and eye aspect ratio.	7
6	PoseTrack	MediaPipe, OpenCV	Tracks body and hand keypoints in 3D space.	6
7	Object Detection (Gesture-Based)	OpenCV, Python	Human detection with gesture-based control interface.	6
8	Hand Gesture Visual Effects	MediaPipe, OpenCV	Gesture-triggered effects for real-time AR/VR experiences.	5
9	Smart Blind Stick	Arduino, Ultrasonic Sensor	Obstacle detection hardware for visually impaired.	4

10	REEGA – Scene-to-Speech AI	YOLOv5, Wikipedia API, TTS	Describes detected scenes verbally using web data.	8
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Machine Learning & Data Analytics

No.	Project	Tech Stack	Description	 Hardness (1–10)
11	Student Performance Classifier	Scikit-learn, Pandas	Predicts student outcomes via ML classification.	5
12	Loan Approval Predictor	Python, Scikit-learn	Classifies loan approvals using banking data.	5
13	Housing Price Predictor	Python, Sklearn, Pandas	Regression model for real estate price prediction.	5
14	Retail Sales Forecasting	Statsmodels, Prophet	Time-series forecasting for retail sales optimization.	7
15	Titanic Survival Predictor	Scikit-learn, Seaborn	Binary classification benchmark project.	3
16	Retail Inventory System	Tkinter, SQLite	CRUD-based desktop system for retail analytics.	4
17	Power BI & Tableau Dashboards (10+)	Power BI, Excel	Created analytics dashboards for HR, sales, students, and finance.	6
18	CSV to Database App	Streamlit, Pandas, SQLite	Converts CSVs into searchable structured data tables.	4
19	SimpleMLTools	NumPy, PIL	Utility toolkit for data preprocessing and image operations.	3



Generative AI & Automation Projects

No.	Project	Tech Stack	Description	 Hardness (1–10)
20	PaperWhisper.AI	Python, LLM API	Research summarizer that converts academic papers into insights.	7
21	AI Data Analyst	Python, OpenAI API	Conversational AI for analyzing datasets and producing insights.	7
22	Vendor-Quotation Bot	Python, FastAPI, LLM	Automates vendor discovery and quotation generation from RFP text.	8
23	JobJockey.AI	Python, LLMs	Reads resumes and generates personalized interview prep and feedback.	7
24	OpenBook Blog Forge	Python, LangChain	AI debater that challenges user opinions and generates blog responses.	8
25	TimeWhisper.AI	Python, Streamlit, LLM	AI-based timesheet automation and reporting tool.	6
26	MentorKrishna.AI	Python, LangChain, LLM	AI therapist bot offering emotional guidance via conversational reasoning.	8
27	LogiQ.AI Chrome Extension	JavaScript, Python API	Chrome assistant extension that integrates GenAI reasoning.	8
28	CodePilot.AI	Python, OpenAI API	Code helper that explains, debugs, and generates optimized snippets.	7

Applications, APIs & Systems

No.	Project	Tech Stack	Description	 Hardness (1–10)
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29	Background Removal API	Flask, DeepLabV3	Web API that removes image backgrounds efficiently.	6
30	Real-Time Image Resizer API	Flask, OpenCV	Lightweight resizing service for web pipelines.	4
31	Retail 3-Way Log System (PAIRS)	Python, Streamlit, SQLite	Tracks and analyzes mall customer behavior and purchase patterns.	8
